Sahaj Bhadja

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Summary

AI-native software engineer with 2+ years shipping production web applications and intelligent workflows using Python, TypeScript, React, Node.js, SQL, and PyTorch. Experienced in LLM systems, distributed services, testing, CI/CD, scalability, and end-to-end ownership.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built production Python APIs and Azure SQL services for 500+ users, optimizing indexed queries, schemas, and execution paths to reduce response latency from 1.2 seconds to 350 milliseconds.
- Restructured a fault-tolerant LLM orchestration system with explicit job states and failure isolation, processing 1,000+ AI jobs while tripling throughput and improving observability, recovery, and reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services handling 10,000+ daily requests, implementing secure document ingestion, authentication, AI grading workflows, and error handling while debugging production issues affecting more than 150 users.
- Shipped React workflows and multithreaded processing for 1,000+ files with AWS RDS persistence, automated validation, progress monitoring, and failure recovery, reducing turnaround 60% under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built production-oriented Python and PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, automated validation, performance profiling, model evaluation, and systematic debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable Python services and data integrations using AWS Glue, S3, SQL, pagination, and caching, improving production response times 30% and processing efficiency 68% across connected applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect investigation, code reviews, regression verification, and reliable CI/CD releases across multiple environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed TypeScript and Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and synchronization, securely processing more than two million Azure SQL records through documented APIs.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Skills

Languages & Web: Python, TypeScript, JavaScript, Java, C++, SQL, React, Next.js, Node.js, Flask, FastAPI, HTML, CSS, REST APIs, WebSockets
AI & Data: LLM APIs, natural language processing, PyTorch, transformers, embeddings, semantic search, FAISS, Pinecone, model evaluation, PostgreSQL, Azure SQL, MongoDB, AWS RDS
Systems & Quality: Distributed systems, microservices, system design, caching, multithreading, automated testing, unit testing, integration testing, PyTest, Mockito, debugging, monitoring, failure recovery
Delivery & Collaboration: Git, GitHub Actions, Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, GCP, Agile, code reviews, technical documentation, production support, AI coding agents

Projects

Big-Brain | Live | *Python, React, PostgreSQL, GCP, REST APIs, AI*
- Built a secure AI-assisted document application over 50,000+ files using React, Python APIs, PostgreSQL, semantic retrieval, optimized indexing, automated validation, and cloud deployment with sub-second search.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built a polished AI-native image platform with Next.js, MongoDB, Stripe, CLIP, and SAM, delivering responsive editing workflows, scalable subscriptions, and production experiences that increased premium conversion 18%.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a client-server platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, WebSockets, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.